

Hashing for Retrieving Long-Tailed Distributed Remote Sensing Images

Lirong Han¹, Student Member, IEEE, Mercedes E. Paoletti², Senior Member, IEEE,
Sergio Moreno-Álvarez³, Graduate Student Member, IEEE, Juan M. Haut⁴, Senior Member, IEEE,
Rafael Pastor-Vargas⁵, Senior Member, IEEE, and Antonio Plaza⁶, Fellow, IEEE

Abstract—The widespread availability of remotely sensed datasets establishes a cornerstone for comprehensive image retrieval within the realm of remote sensing (RS). In response, the investigation into **hashing-driven** retrieval methods garners significance, enabling proficient image acquisition within such extensive data magnitudes. Nevertheless, the used datasets in practical applications are invariably less desirable and with **long-tailed distribution**. The primary hurdle pertains to the substantial discrepancy in class volumes. Moreover, commonly utilized RS datasets for hashing tasks encompass approximately two–three dozen classes. However, real-world datasets exhibit a randomized number of classes, introducing a challenging variability. This article proposes a **new centripetal intensive attention hashing (CIAH) mechanism based on intensive attention features for long-tailed distribution RS image retrieval**. Specifically, an intensive attention module (IAM) is adopted to enhance the significant features to facilitate the subsequent generation of representative hash codes. Furthermore, to deal with the inherent imbalance of long-tailed distributed datasets, the utilization of a centripetal loss function is introduced. This endeavor constitutes the inaugural effort toward long-tailed distributed RS image retrieval. In pursuit of this objective, a collection of long-tail datasets is meticulously curated using four widely recognized RS datasets, subsequently disseminated as benchmark datasets. The selected fundamental datasets contain **7, 25, 38, and 45** land-use classes to mimic different real RS datasets. Conducted experiments demonstrate that the proposed methodology attains a performance benchmark that surpasses currently existing methodologies.

Index Terms—Attention mechanisms, centripetal loss, hashing, long-tailed remote sensing (RS), RS image retrieval.

Manuscript received 1 September 2023; revised 14 December 2023; accepted 9 January 2024. Date of publication 31 January 2024; date of current version 15 February 2024. This work was supported in part by the Consejería de Economía, Ciencia y Agenda Digital of the Junta de Extremadura and the European Regional Development Fund (ERDF) of the European Union under Grant GR21040; in part by the ERDF of the European Union Interreg V-A España-Portugal [Programa de Cooperación Transfronteriza Interreg España-Portugal (POCTEP)] 2021-2027 Program, under Grant 0206_RAT_EOS_PC_6_E; in part by the Spanish Ministerio de Ciencia e Innovación under Project PID2019-110315RB-I00 (APRISA); and in part by the UNED. (Corresponding author: Rafael Pastor-Vargas.)

Lirong Han, Mercedes E. Paoletti, Juan M. Haut, and Antonio Plaza are with the Department of Technology of Computers and Communications, University of Extremadura, 10001 Cáceres, Spain (e-mail: lironghan_upc@163.com; mpaoletti@unex.es; juanmariohaut@unex.es; aplaza@unex.es).

Sergio Moreno-Álvarez is with the Department of Computer Systems and Languages, National University of Distance Education, 28015 Madrid, Spain (e-mail: smoreno@lsi.uned.es).

Rafael Pastor-Vargas is with the Departamento de Sistemas de Comunicación y Control, Escuela Técnica Superior de Ingeniería Informática, Universidad Nacional de Educación a Distancia (UNED), 28040 Madrid, Spain (e-mail: rpastor@scc.uned.es).

Digital Object Identifier 10.1109/TGRS.2024.3360621

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See <https://www.ieee.org/publications/rights/index.html> for more information.

I. INTRODUCTION

DUE to the rapid development of **remote sensing** (RS) technology in recent years, the volume of information has expanded dramatically. The intensification of data acquisition missions significantly impacts the surge in data volume [1], [2], [3]. Obtained images are procured through a diverse array of modalities, including **optical images (OIs)**, as well as **multi-spectral (MSIs) and hyperspectral (HSIs) scenes**. These images have transcended their role to become indispensable tools within a large range of disciplines. Particularly, they assume a central role in disaster response and recovery operations, urban planning, or land-use analysis. In the realm of disaster response, RS imagery bestows crucial insights essential for evaluating and managing catastrophic situations [4], [5], [6]. In urban planning, this imagery provides decision-makers with invaluable data to foster sustainable urban development and optimize infrastructural configurations [7], [8], [9]. In the context of land-use analysis, these data substantially contribute to the elucidation of patterns, thereby amplifying endeavors related to resource management and environmental monitoring [10], [11], [12]. A plethora of applications and tasks derive benefits from extensive and information-rich RS databases. Within the framework of image retrieval, an imperative challenge arises: the adept and efficient retrieval of images pertinent to a specific situation. In the event of disasters, **fast retrieval** of required images from real-time acquisitions is paramount for swift disaster rescue operations. Indeed, through the analysis of images depicting disaster scenes, rescuers can formulate timely and precise rescue strategies. In this context, the methodology of hashing has attracted the attention of scientific researchers around the world.

In this regard, **hashing** is a conventional retrieval technique expounded in relevant literature [13], [14], [15], [16], [17]. It involves the quantization of initial data into a binary code of fixed length, denoted as a hash code. The objective underlying hashing techniques reside in the utilization of low-dimensional hash codes instead of the original data. Thus, the fundamental principle revolves around preserving a consistent approximation of the near-neighbor relationships between the initial data and its corresponding representation across distinct spaces. In other words, the near neighboring data in the original spatial space after quantization also maintain the near neighboring relationship in the Hamming space and vice versa. An effective hashing scheme is characterized by the generation of hash

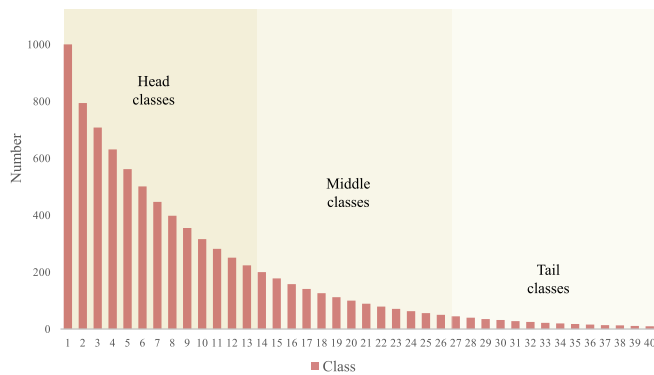


Fig. 1. Example of a long-tailed distribution of samples in a dataset with 40 classes. All classes can be evenly divided into head class, middle class, and tail class. A few head classes occupy most of the samples in the whole dataset. Meanwhile, a larger number of middle and tail classes occupy only a small number of samples.

codes that encapsulate a substantial portion of the essential information present in the original data, i.e., features, while concurrently upholding discriminative attributes. As a consequence, hash codes occupy less memory than the original data. Furthermore, the **Hamming distance** between hash codes can be quickly obtained by simple matrix operations when performing retrieval. With the benefit of low storage and fast computation, hashing has been extensively exploited in the RS image retrieval field [18], [19], [20], [21], [22].

A. **RS Hashing**: Methods and Boundaries

Consistent with diverse methods of feature extraction, hash techniques can be categorized into shallow and deep hashing. The former refers to the feature extraction of original data using traditional feature extractors, e.g., SIFT [23] or GIST [24]. In such methods, the predesigned hash function is utilized to map the extracted shallow features into hash codes. There are several studies that have been devoted to introduce shallow hashing to the RS image retrieval community. For instance, Demir and Bruzzone [25] introduce two kernel-based nonlinear hashing methods (unlabeled and labeled) into RS image retrieval to speed up the retrieval process. Conversely, an alternative investigation [26] introduces a partially random hash approach that enhances the efficiency of hash function construction. This procedure involves the random initialization of specific hash learning parameters, while the remaining parameters are derived through data-driven learning. In addition, Reato et al. [27] extract feature clusters of RS images and generate unique hash codes for each individual cluster. In this regard, employing multiple hash codes for image representation contributes to an improvement in the accuracy of retrieval outcomes. Finally, Fernandez-Beltran et al. [28] propose a probabilistic latent semantic model to mine the potential semantic information of RS images while preserving the information within the hash code. Notwithstanding the structural simplicity and rapid execution, the efficacy of shallow hashing is constrained by the capabilities of conventional feature extractors, leading to unsatisfactory retrieval.

The remarkable feature extraction capabilities demonstrated by deep learning [29], [30], [31], [32], [33] have prompted

a growing interest in deep-learning-based hashing, i.e., deep hashing. Thus, some efforts have been conducted to incorporate these methodologies into the field of RS image retrieval. Concretely, Li et al. [34] delve into the feasibility of deep hashing, yielding superior performance. Liu et al. [35] aim to integrate the multiscale features and use the attention map to increase the retrieval precision. Song et al. [36] propose a novel hashing model that integrates both hashing and classification loss functions for simultaneous model optimization. As a novelty, Song et al. [37] introduce an asymmetrical approach to hash code learning. This methodology employs distinct hashing methods to generate hash codes for queries and database entries, respectively. The objective is to retain a significant portion of the fundamental information throughout this process. Additionally, numerous studies have been conducted on the basis of fast hashing retrieval advantages. Concretely, the investigation [38] involves the quantification of local patches within extensive RS images into hash codes. This methodology enables a fast target localization in large-scale scenes. Similarly, Han et al. [39] design an encryption scheme based on hashing to counteract RS image target localization.

Although the aforementioned deep hashing approaches have demonstrated satisfactory retrieval capabilities, existing image hashing methods are based on predesigned RS datasets. This fact refers to a dataset in which the classes and the samples contained in each class are preset to an optimal state, e.g., the classes are set to be between 20 and 30, and the number of samples in each class is approximately **balanced**. However, the distribution inherent to actual real-world scenarios does not align with these ideal expectations, quite the contrary. This is shown in Fig. 1 for a **long-tailed class distribution**.

B. Problem Description

The abundance of existing RS image data, particularly in optical imagery, offers a robust foundation for the pursuit of large-scale image retrieval within the ambit of Big Data. Nonetheless, real-world **optical RS databases** encounter a significant challenge stemming from **imbalanced** class data. The notion of imbalance pertains to the disparate distribution of samples among classes, leading to uneven quantities for each. For instance, within a database containing urban scenes, the count of building samples markedly surpasses the greenery samples. Conversely, in the countryside, the prevalence of farmland samples outweighs that of residential housing samples. Regrettably, the majority of prevailing RS image retrieval studies are conducted under the assumption of an ideal state, wherein the volume of data for each class is either equivalent or closely matched. However, achieving such conditions in real-world applications presents considerable challenges.

Remotely sensed data manifest a long-tailed class distribution, a characteristic extensively discussed in [40], [41], [42], and [43], where a limited set of sample classes dominates the dataset capacity, and the remaining large number of classes occupy only a small amount. In conjunction with this challenge, the majority of prevailing deep hashing methodologies are trained utilizing the tuple similarity loss function. This loss function is based on a set of samples and the similarity

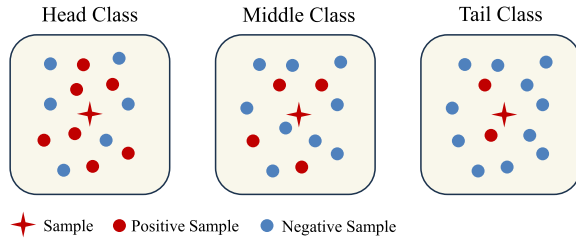


Fig. 2. Unbalanced long-tailed dataset. Stars denote one sample, dots with the same color (red) denote samples belonging to the same class, and different colors (blue) denote samples belonging to different classes. Within the head class, a sample is characterized by a profusion of positive instances. In the intermediate class, a sufficient number of positive instances is observed. Conversely, the tail class is marked by a scarcity of positive instances for a given sample.

relationship between them. During the training process, the network receives input in the form of tuple samples and subsequently generates hash codes that are strategically positioned in proximity from each other, guided by the underlying similarity relationship. However, in long-tailed distributed datasets, the number of similar samples in the middle and tail classes is significantly less than the number of dissimilar samples, as shown in Fig. 2. This causes that the samples in the middle and tail classes have minimal contribution to the model parameters during training. The models concentrate their attention on the head class samples, thus leading to the attenuation of the overall performance. Unfortunately, with such a realistic issue, no antecedent research endeavors have been undertaken in the field of RS for long-tailed distributed image retrieval.

C. Novel Solution: CIAH

In response to the existing research gap and with the goal of enhancing the applicability of RS hashing within real-world scenarios, a novel methodology is introduced, i.e., the “centripetal intensive attention hashing” (CIAH). The proposed model addresses the need for efficient retrieval of long-tailed RS image datasets. Initially, an intensive attention module (IAM) is incorporated to enhance preextracted features, thereby retaining essential information. Subsequently, to mitigate the challenges arising from the pronounced class imbalance inherent to long-tailed data, a centripetal loss function is employed to optimize the model. The hash code of each sample is as close as possible to its predefined hash code center. Hence, there is no consideration of the imbalance between the number of similar and dissimilar sample tuples during training. Conclusively, the absence of datasets tailored for the retrieval of long-tailed distributed RS images stems from the lack of attention directed toward this specific task. In light of this, an initiative is undertaken to create a collection of long-tailed RS datasets encompassing three shrinkage scales. These datasets are derived from four widely recognized RS datasets, resulting in a total of 12 distinct datasets. This compilation stands as a benchmark resource.

D. Contributions of This Work

The primary innovative facets of this article are summarized through the presentation of the following contributions.

- 1) To the best of authors’ knowledge, this work represents the first attempt in the RS community to address long-tailed distributed datasets, aiming to produce an image retrieval experimental environment that closely mirrors real-world conditions.
- 2) In order to mitigate the problems posed by data imbalance, two strategies are implemented. First, an IAM is exploited to enhance the features. Second, the proposed model is optimized using a centripetal loss function. These two tactics significantly improve the proposal to effectively exploit the long-tailed distributed RS image retrieval.
- 3) A comprehensive collection of long-tailed RS datasets is established, leveraging four prominent datasets with three different shrinkage scales. To foster further exploration in this research domain, these datasets are made publicly accessible, thereby serving as a benchmark resource for the corresponding task.

The remainder of this article is organized as follows. Section II describes the designed methodology, including the IAM and the centripetal loss function. Section III explains the environment and the experimental setting. In addition, it provides the experimental results and the analysis. Finally, Section IV summarizes the contributions and findings of this article.

II. METHODOLOGY: CIAH

The initial phase involves the modeling of the hashing process, elucidating the essential import of primary symbols and functions. Subsequently, the architecture of the proposed CIAH is expounded upon. This is followed by the introduction of the intensive attention model. Lastly, a comprehensive elucidation of the centripetal loss function is provided.

A. Hash Modeling

The image dataset is defined as $X = \{x_1, x_2, \dots, x_n\}_{n=1}^N$, and its corresponding label set and hash code set are $L = \{l_1, l_2, \dots, l_n\}_{n=1}^N$ and $H = \{h_1, h_2, \dots, h_n\}_{n=1}^N$, respectively, where N denotes the number of samples. In $l_n \in \{1, 2, \dots, C\}$, C is the number of classes, while in $h_n \in \{0, 1\}^K$, K is the hash code length. As the number of samples in each class in the long-tailed dataset is inconsistent, the volume of samples in each class is denoted by $r = \{r_1, r_2, \dots, r_C\}$. Among these, r_1 represents the highest count of samples allocated to the initial class, with the numbers of samples progressively decreasing across the remaining classes. The first third of the classes is defined as the head class, the middle third as the middle class, and the last third as the tail class. The setting of r will be described in detail in the dataset description in future experiments.

In this context, the proposed CIAH has four components: 1) feature extraction backbone $\mathcal{F}_\Omega(\cdot)$; 2) IAM $\mathcal{M}_{\Omega_m}(\cdot)$; 3) hash learning layer $\mathcal{M}_{\Omega_h}(\cdot)$; and 4) classifier layer $\mathcal{M}_{\Omega_c}(\cdot)$. These four components are illustrated in Fig. 3. The Ω represents the model parameters. The overall network can be regarded as a hash function $\mathcal{H}(\cdot)$, indeed. Thus, the process of hash code learning is denoted by $H = \mathcal{H}(X)$.

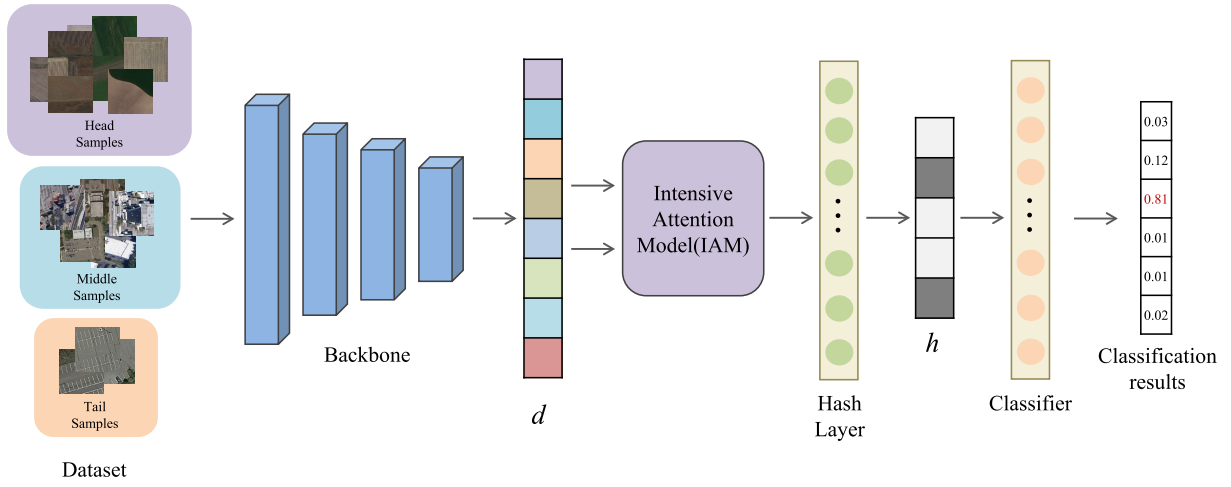


Fig. 3. Flowchart of the proposed CIAH. The proposed model consists of four parts: a backbone, an IAM, a hash learning layer, and a classifier layer. Specifically, the backbone is engaged for the extraction of features, while the IAM serves to enhance these features. The hash learning and the classifier layers are responsible for generating hash codes and producing classification outcomes, respectively.

B. Delving Into the Architectural Design of CIAH

The subsequent stage involves the elucidation of the structure and operational attributes pertaining to the aforementioned four components within the architectural framework of CIAH. Moreover, a backbone is employed to extract preliminary features, with these features denoted as d . To expedite and optimize the feature extraction process, the widely utilized ResNet-34 architecture [44] is adopted as the backbone network given its shallow architecture and remarkable proficiency. Then, the IAM is introduced to enhance the preliminary extracted features, which are used for subsequent hash code learning and classification. Both the hash and classifier layers contain a fully connected layer and a tanh activation function. The hash layer generates hash codes from the enhanced features. Then, the hash codes are used as inputs to the classifier layer, the outputs of which are classification results. Therefore, the generation of hash codes is constrained by both the hashing task and the classification task.

Given the significant class imbalance inherent to the long-tailed dataset, the trained model tends to prioritize the head category, while relatively neglecting the tail category. To address this limitation, two strategies are implemented: 1) the IAM module and 2) a centripetal loss function. In this context, the IAM enhances the preliminary features obtained, and the centripetal loss function is used to increase the importance of the tail classes during training. These strategies are further elucidated in the subsequent stages.

C. Intensive Attention Mode

Hashing is dedicated to quantize high-dimensional data into a low-dimensional hash code. During retrieval, the hash code replaces the original data to perform the retrieval. Evidently, information loss during the dimensionality reduction process is an inherent phenomenon. To bring the hash code in closer alignment with the original data, the initial step entails acquiring features that encapsulate more information. Consequently, the IAM proposal is introduced to amplify these features, as illustrated in Fig. 4.

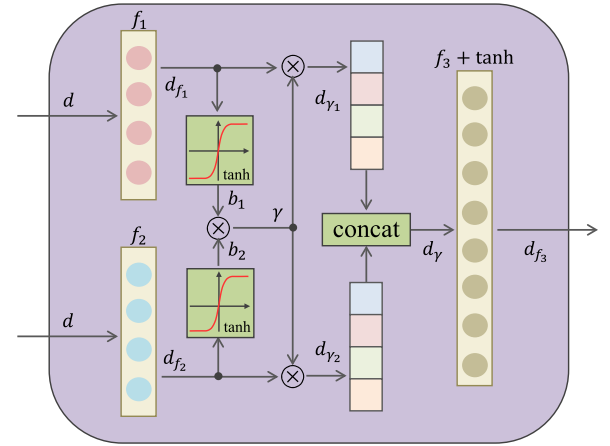


Fig. 4. IAM.

First, the preliminary features d extracted by the backbone are input to the IAM to produce the output enhanced features d_{f_3} . The IAM consists of three fully connected layers (f_1 , f_2 , and f_3) and their corresponding weights ($W_1 = R^{\ell_d \times \ell_{d_{f_1}}}$, $W_2 = R^{\ell_d \times \ell_{d_{f_2}}}$, and $W_3 = R^{\ell_{d_y} \times \ell_{d_{f_3}}}$). Identify d_{f_1} , d_{f_2} , and d_{f_3} as the output features of f_1 , f_2 , and f_3 , respectively, and ℓ_d , $\ell_{d_{f_1}}$, $\ell_{d_{f_2}}$, and $\ell_{d_{f_3}}$ as the dimensions of feature d , d_{f_1} , d_{f_2} , and d_{f_3} , respectively. Also, γ indicates the attention parameter, d_γ is the feature added to the attention, and ℓ_{d_γ} represents the dimension of d_γ . Simultaneously, f_1 and f_2 have the same output length. Therefore, d_{f_1} and d_{f_2} can be formulated as shown in the following equation:

$$d_{f_1} = dW_1, \quad d_{f_2} = dW_2. \quad (1)$$

In the initial state, f_1 and f_2 share the same parameters, and thus, d_{f_1} and d_{f_2} are the same at this stage. In this situation, tanh is set as the activation function for d_{f_1} and d_{f_2} . Concretely, for the hashing task, the preferred operation is to turn d_{f_1} and d_{f_2} into binary vectors using the sign function. Nevertheless, the involvement of the sign function within the network back-propagation process tends to accentuate the

attention, thereby potentially neglecting the middle attention weights. After that, the activated binary-like features b_1 and b_2 are multiplied to obtain the attention parameter γ , as shown in the following equations:

$$b_1 = \tanh(d_{f_1}), \quad b_2 = \tanh(d_{f_2}) \quad (2a)$$

$$\gamma = b_1 b_2. \quad (2b)$$

Then, d_{γ_1} and d_{γ_2} are obtained by adding attention parameter γ to d_{f_1} and d_{f_2} . Finally, d_{γ_1} and d_{γ_2} are concatenated to generate the enhanced feature d_γ , in accordance with the formulation of the following equation:

$$d_\gamma = \text{concat}(d_{\gamma_1}, d_{\gamma_2}). \quad (3)$$

The concat notation refers to the concatenate operation. Finally, a fully connected layer f_3 is added in the following equation to generate the output of IAM:

$$d_{f_3} = d_\gamma W_3. \quad (4)$$

The influence attributed to the newly introduced attention parameter γ is clarified through the following elucidation. Given the initial configuration (where parameters for f_1 and f_2 are shared), the similarity between d_{f_1} and d_{f_2} remains consistent throughout the training process. This results in the values of b_1 and b_2 having the same sign at most positions. Therefore, the values of γ are majority positive and few values are negative. In summary, γ increases the attention to features with the same sign, facilitating the subsequent generation of discriminative hash codes.

D. Centripetal Loss Function

In the context of existing deep hashing methodologies, the prevailing approach involves the training of models through the utilization of a tuple similarity loss function. This loss function integrates multiple samples (along with their respective similarity relationships) as essential inputs. The overarching objective is to minimize the distance between hash codes of similar samples, while maximizing the distance between hash codes of dissimilar samples. Concretely, pairwise loss and triplet loss are popular tuple similarity losses. The former takes two samples as a group and sets their relationship as similar or dissimilar, and the latter delves into the systematic organization of three distinct samples within a group, encompassing the samples themselves, the samples that exhibit positive correlation, i.e., similarity, and those that demonstrate negative correlation, i.e., dissimilarity. Unlike pairwise loss, triplet loss considers both the distance of similar hash codes and dissimilar hash codes.

Nevertheless, the tuple similarity loss function is not applicable in long-tailed distribution datasets due to the limited presence of similar samples within the middle and tail class samples. Thus, this discrepancy induces an imbalance during the training process, prompting the model to disproportionately prioritize the head classes.

To address this limitation, a loss function based on the predetermined hash code center is proposed. This approach, known as the centripetal loss, aims to cluster the hash codes of all samples around this designated center [45]. Centripetal

Algorithm 1 Pseudo-Code of the CIAH Training Process

Input: A pre-trained feature extraction backbone with parameters $\mathcal{F}_\Omega(\cdot)$, A training image dataset with N samples x_n , hyperparameter γ , W_1 , W_2 and W_3 in the IAM $\mathcal{M}_{\Omega_m}(\cdot)$, hash learning layer $\mathcal{M}_{\Omega_h}(\cdot)$, classifier layer $\mathcal{M}_{\Omega_c}(\cdot)$.

Output: The proposed CIAH model.

```

1: for  $n = 1$  to  $N$  do
2:    $d = \mathcal{F}_\Omega(x_n)$ 
3:    $d_\gamma = \mathcal{M}_{\Omega_m}(d)$ 
4:    $h_n = \mathcal{M}_{\Omega_h}(d_\gamma)$ 
5: end for
6: Compute the hash code center of each class
    $\{h_o^1, h_o^2, \dots, h_o^n\}$ 
7: for each batch  $\mathcal{X}$  do
8:    $d_{f_1} = W_1 \mathcal{X}, d_{f_2} = W_2 \mathcal{X}$ 
9:    $b_1 = \tanh(d_{f_1}), b_2 = \tanh(d_{f_2})$ 
10:   $\gamma = b_1 \cdot b_2$ 
11:   $d_\gamma = \text{concat}(d_{\gamma_1}, d_{\gamma_2})$ 
12:   $d_{f_3} = d_\gamma W_3$ 
13:   $h_{\mathcal{X}} = \mathcal{M}_{\Omega_h}(d_{f_3})$ 
14:   $\hat{y} = \text{softmax}(\mathcal{M}_{\Omega_c}(h_{\mathcal{X}}))$ 
15:  Compute  $\mathcal{L}_{C_1}$  and  $\mathcal{L}_{C_2}$ 
16:   $\mathcal{L} = \mathcal{L}_{C_1} + \beta \mathcal{L}_{C_2}(\hat{l}, l)$ 
17:  Optimizing the network parameters based on  $\mathcal{L}$ 
18: end for
19: return CIAH

```

loss simply serves to concentrate on the compactness of the samples in relation to their corresponding centers, notably alleviating the imbalance caused by tuple loss. First, the process defines the center of the predetermined hash code, as outlined in the following equation:

$$h_o^i = \frac{1}{r_i} \sum_{n=1}^{r_i} h_n. \quad (5)$$

Here, h_o^i represents the center of the i th class, and r_i indicates the volume of samples belonging to this class. Additionally, h_n refers to the hash code obtained without training, solely relying on the pretrained model. Then, the hash code center set $H_o = \{h_o^1, h_o^2, \dots, h_o^C\}$ is obtained.

Consequently, the adopted hash center encompasses the original features of the data. During training, the objective is for the hash code of each sample to approximate its corresponding class center as closely as possible. The Hamming distance between the hash code h_n and its corresponding class center h_o^i is mathematically represented by the following equation:

$$D(h_n, h_o^i) = \frac{1}{2} (K - h_n^T h_o^i). \quad (6)$$

However, given that the Hamming distance is not differentiable, an alternative approach becomes necessary. In this regard, the cosine distance is proposed. The calculation of the cosine distance between two hash vectors h_n and h_m can be represented through the following equation:

$$\cos \theta_{nm} = \frac{h_n^T h_m}{\|h_n\| \cdot \|h_m\|}. \quad (7)$$

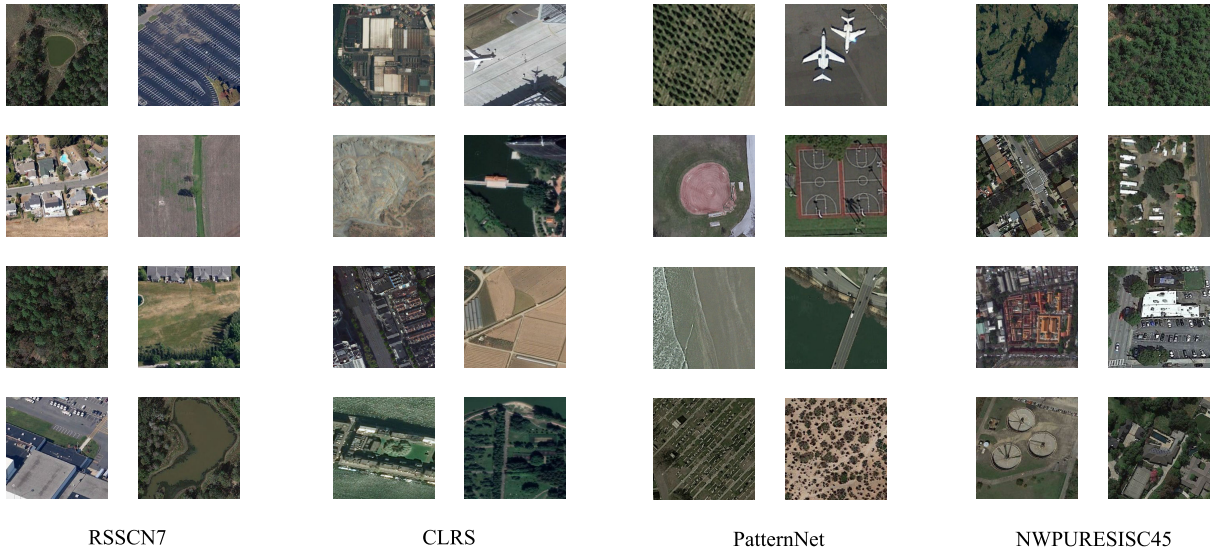


Fig. 5. Representative samples of RSSCN7, CLRS, PatternNet, and NWPURESISC45 datasets.

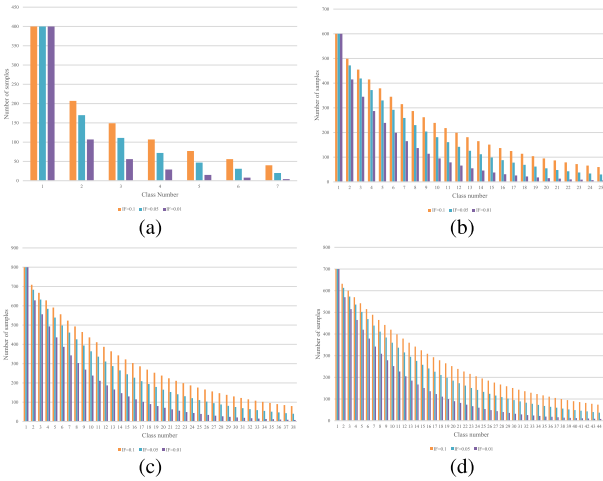


Fig. 6. Class data distribution of the produced long-tail RS dataset. (a) RSSCN7 dataset. (b) CLRS dataset. (c) PatternNet dataset. (d) NWPURESISC45 dataset.

Since the cosine distance is differentiable and inversely proportional to the Hamming distance, the centripetal loss $\mathcal{L}_{C_1}$ is shown as follows:

$$\mathcal{L}_{C_1} = -\log \frac{\exp(\partial \Theta \langle h_n, h_o^i \rangle)}{\sum_{j \in 1, C, j \neq i} \exp(\partial \Theta \langle h_n, h_o^j \rangle)} \quad (8)$$

where $\Theta(\cdot)$ denotes the cosine distance, and ∂ signifies the relevant importance parameters. As depicted in (8), it becomes evident that the centripetal loss enforces the proximity of hash codes to the corresponding centers. In conjunction with the centripetal loss, an additional classification loss is incorporated to generate more discriminative hash codes. As such, the comprehensive loss function $\mathcal{L}$ governing the entire network is defined through the expression presented in the following equation:

$$\mathcal{L} = \mathcal{L}_{C_1} + \alpha \mathcal{L}_{C_2}(\hat{l}, l). \quad (9)$$

The $\mathcal{L}_{C_2}$ represents the classification loss, $\hat{l}$ denotes the predicted label, and α is the weight parameter. By optimizing (9), the CIAH is trained to produce discriminative hash codes. In order to demonstrate CIAH clearly, Algorithm 1 provides the pseudo-code of the CIAH training process.

III. EXPERIMENTAL RESULTS

A. **Datasets** Description

To the best of authors' knowledge, no publicly available long-tail RS dataset has been established previously. Therefore, several long-tail RS datasets have been designed, drawing from four widely used classical RS datasets.

1) **RSSCN7**: The RSSCN7 dataset [46] was collected from Google Earth and released in 2015. It contains 2800 OIs of seven representative land-use classes, including grassland, forest, farmland, parking lot, residential area, industrial area, and river and lake. Each of these classes contains 400 images, all with 400×400 pixels. These samples are sampled based on four different shrinkage scales, with a resolution of 2.7–20 m.

2) **CLRS**: The CLRS dataset [47] was collected from Google Earth, BingMap, GoogleMap, and Tiand and released in 2020. It includes 25 land-use classes with 600 images per class, for a total of 15 000 OIs. Each image has a size of 256×256 pixels and a resolution of 0.26–8.85 m.

3) **PatternNet**: The PatternNet dataset [48] was collected from Google Earth, too, and published by Wuhan University, Wuhan, China, in 2017. It includes 38 commonly used land-use classes with 800 OIs per class, for a total of 30 400 samples. The size of each image is 256×256 pixels, and the resolution is 0.06–4.7 m.

4) **NWPURESISC45**: The NWPURESISC45 dataset [49] was created by Northwestern Polytechnic University, Xi'an, Shaanxi, China, in 2016 based on Google Maps imagery. It contains a total of 31 500 images of 256×256 pixels, covering 45 classes with 700 images per class and resolution of 0.2–3 m.

TABLE I

SAMPLE VOLUMES OF THE TEST AND TRAINING SETS FOR
FOUR LONG-TAIL RS DATASETS (RSSCN7, CLRS,
PATTERNNet, AND NWPURESISC45)

Dataset	RSSCN7			CLRS		
	IF=0.01	IF=0.05	IF=0.1	IF=0.01	IF=0.05	IF=0.1
All	619	876	1036	3037	4544	5649
Test	100	100	100	500	500	500
Train	519	776	936	2537	4044	5149
Dataset	PatternNet			NWPURESISC45		
	IF=0.01	IF=0.05	IF=0.1	IF=0.01	IF=0.05	IF=0.1
All	6242	9326	11575	6503	9709	12037
Test	1000	1000	2000	1000	1000	2000
Train	5242	8326	9575	5502	8709	10037

Fig. 5 shows some representative samples of the above four RS datasets. In order to delve into the retrieval of long-tailed distributed RS images, three distinct datasets (each exhibiting varying levels of shrinkage) have been crafted for every RS dataset, in accordance with the following equation:

$$T_i = \begin{cases} \max, & i = 1 \\ \max \cdot \text{IF}^{\frac{i}{C}}, & i = 2, \dots, C \end{cases} \quad (10)$$

where T_i denotes the sample number of the i th class, max and min denote the maximum and minimum volumes of the number of samples from each class, respectively, and IF represents the shrinkage scale, which is defined as $\text{IF} = (\min / \max)$ and takes values of 0.1, 0.05, and 0.01. The class data distribution of the produced long-tail RS dataset is shown in Fig. 6. From now on, a designated long-tailed RS dataset is denoted as: dataset name (IF value), e.g., PatternNet (IF = 0.01).

The aim is to ensure that all available classes are represented in the dataset, capturing the diversity of RS image scenarios. However, tail classes may have fewer samples than head classes due to the inherent characteristics of the long-tailed distribution. To ensure fairness and reduce bias in the dataset creation process, random sampling techniques are adopted. This entailed randomly selecting samples within each class to establish a representative long-tailed distribution. Since the volume of each long-tail dataset is different, the volumes of the test and training sets for each dataset are provided in Table I. Furthermore, the above long-tail data named LTRSdataset have been made publicly available.¹

B. Experimental Settings

A detailed exposition of the evaluation metrics and parameter settings is presented in the following. Moreover, a diverse set of well-established evaluation metrics is employed to comprehensively assess the retrieval efficacy. This underscores the systematic approach to evaluate the proposed method and highlights the considerations that parameter adjustments during experimentation inform.

¹https://drive.google.com/file/d/1lhFJfBjSkwTQaYzTOdt4i-NiYjqCIhoq/view?usp=drive_link

1) *Evaluation Metrics*: The evaluation metrics for hash retrieval tasks can be categorized into numerical and visual metrics. Numerical metrics are precision, recall, and mean average precision (mAP). Meanwhile, visual metrics are the precision–recall curve and the t-distributed stochastic neighbor embedding (t-SNE) distribution. Precision, as defined in the following equation, represents the percentage of correct results among all retrieved results, denoted as $\text{All}_{\text{results}}$:

$$\text{Precision} = \frac{\text{True}}{\text{All}_{\text{results}}}. \quad (11)$$

On the other hand, recall is defined in the following equation as the percentage of correct results among all correct results, denoted as All_{true} :

$$\text{Recall} = \frac{\text{True}}{\text{All}_{\text{true}}}. \quad (12)$$

Finally, the mAP can be calculated as shown in the following equation:

$$\text{mAP} = \frac{1}{Q} \sum_{q=1}^Q \frac{1}{R} \sum_{r=1}^R \text{Precision}(r) \text{true}(r) \quad (13)$$

where Q denotes the number of query samples and R denotes the number of retrieval results. $\text{Precision}(r)$ is the precision of Top- r , $\text{true}(r) = 1$ when the retrieval result of rank r is correct, and 0 otherwise. Specifically, mAP is not only concerned with the precision of retrieval results but is also related to the order of retrieval. Higher mAP values correspond to accurately retrieved results with top rankings, while lower values indicate otherwise. Given the widespread adoption of mAP as a standard metric, it is exclusively presented as the sole numerical measure in the forthcoming experimental results. Precision and recall are visually represented as plotted curves. Moreover, the precision–recall curve gives a straightforward indication of the performance of the method. A larger area under this curve identifies an improved retrieval performance. Finally, t-SNE (as a visual dimensionality reduction method) reduces high-dimensional data to two or three dimensions and demonstrates its approximate nearest-neighbor distances. Since the central aim of the hashing task revolves around preserving close hash code proximity for similar samples and ensuring separation for hash codes of dissimilar samples, t-SNE stands as a suitable visual evaluation metric for comprehensively appraising the hashing performance.

2) *Hardware and Parameter Settings*: The experiments are conducted using a computer environment with Python 3.7.12 and the PyTorch 1.8.0 framework on a GeForce GTX 1660 Ti graphics processing unit (GPU). The learning rate is set to $1e^{-5}$; the batch size is 32, and the number of epochs is 150. The length of the hash code is set to 16, 32, or 64 bits. Finally, the weight parameter α is set to 0.2 according to the ablation study.

C. Results: Analysis and Discussion

Comparative experiments are conducted between CIAH, six competitive common hashing methods and three RS hashing methods, as outlined next: 1) deep supervised hashing

TABLE II
MAP RESULTS ON RSSCN7 LONG-TAILED RS DATASET WITH 16-, 32-, AND 64-BIT CODE LENGTH

	Methods	DHN	DSH	DTSH	DPSH	HashNet	FAH	DHCNN	AHCL	CSQ	CIAH (Ours)
16 bits	RSSCN7 (IF=0.01)	0.9540	0.9575	0.9484	0.9546	0.9353	0.5231	0.8985	0.8532	0.9671	0.9648
	RSSCN7 (IF=0.05)	0.9758	0.9743	0.9724	0.9748	0.9559	0.8219	0.9625	0.9428	0.9674	0.9912
	RSSCN7 (IF=0.1)	0.9703	0.9827	0.9804	0.9743	0.9518	0.7884	0.9627	0.9623	0.9736	0.9891
32 bits	RSSCN7 (IF=0.01)	0.9624	0.9711	0.9448	0.9559	0.9562	0.6985	0.8768	0.7473	0.9755	0.9762
	RSSCN7 (IF=0.05)	0.9772	0.9801	0.9670	0.9738	0.9638	0.8219	0.9625	0.9450	0.9752	0.9794
	RSSCN7 (IF=0.1)	0.9670	0.9797	0.9817	0.9646	0.9668	0.8560	0.9659	0.9747	0.9735	0.9999
64 bits	RSSCN7 (IF=0.01)	0.9380	0.9793	0.9641	0.9579	0.9667	0.5737	0.8645	0.8202	0.9671	0.9755
	RSSCN7 (IF=0.05)	0.9758	0.9557	0.9720	0.9661	0.9662	0.8440	0.9074	0.8872	0.9719	0.9862
	RSSCN7 (IF=0.1)	0.9752	0.9860	0.9757	0.9721	0.9746	0.8657	0.9499	0.9352	0.9818	0.9944

TABLE III
MAP RESULTS ON CLRS LONG-TAILED RS DATASET WITH 16-, 32-, AND 64-BIT CODE LENGTH

	Methods	DHN	DSH	DTSH	DPSH	HashNet	FAH	DHCNN	AHCL	CSQ	CIAH (Ours)
16 bits	CLRS (IF=0.01)	0.8066	0.8812	0.8722	0.8364	0.8483	0.8160	0.8499	0.8770	0.8866	0.9302
	CLRS (IF=0.05)	0.7843	0.8849	0.8718	0.8163	0.8608	0.8158	0.8437	0.8549	0.8883	0.9364
	CLRS (IF=0.1)	0.7297	0.8575	0.8541	0.7769	0.7294	0.7931	0.7897	0.8224	0.8650	0.9008
32 bits	CLRS (IF=0.01)	0.8089	0.8805	0.8722	0.8542	0.8730	0.8126	0.8601	0.8925	0.9037	0.9303
	CLRS (IF=0.05)	0.7835	0.8804	0.8760	0.8596	0.8904	0.8386	0.8727	0.8959	0.8984	0.9370
	CLRS (IF=0.1)	0.7226	0.8738	0.8745	0.8375	0.8681	0.8203	0.8430	0.8775	0.8792	0.9087
64 bits	CLRS (IF=0.01)	0.8289	0.8841	0.8763	0.8532	0.8963	0.8211	0.8696	0.8829	0.9037	0.9367
	CLRS (IF=0.05)	0.7972	0.8867	0.8918	0.8502	0.8608	0.8542	0.8756	0.8762	0.8936	0.9299
	CLRS (IF=0.1)	0.7430	0.8827	0.8737	0.8290	0.8819	0.8268	0.8521	0.8967	0.8870	0.9199

TABLE IV
MAP RESULTS ON PATTERNNET LONG-TAILED RS DATASET WITH 16-, 32-, AND 64-BIT CODE LENGTH

	Methods	DHN	DSH	DTSH	DPSH	HashNet	FAH	DHCNN	AHCL	CSQ	CIAH (Ours)
16 bits	PatternNet (IF=0.01)	0.8795	0.9940	0.9845	0.8634	0.9915	0.9300	0.9889	0.9152	0.9932	0.9995
	PatternNet (IF=0.05)	0.7929	0.9942	0.9770	0.7729	0.9773	0.9659	0.9863	0.8723	0.9931	0.9959
	PatternNet (IF=0.1)	0.8050	0.9989	0.9854	0.7809	0.9659	0.9823	0.9951	0.8269	0.9962	0.9972
32 bits	PatternNet (IF=0.01)	0.9436	0.9972	0.9822	0.9643	0.9847	0.9302	0.9925	0.9670	0.9970	0.9990
	PatternNet (IF=0.05)	0.9274	0.9955	0.9921	0.9480	0.9944	0.9661	0.9875	0.9756	0.9956	0.9963
	PatternNet (IF=0.1)	0.9538	0.9973	0.9962	0.9247	0.9964	0.9689	0.9951	0.9962	0.9954	0.9978
64 bits	PatternNet (IF=0.01)	0.9683	0.9970	0.9907	0.9701	0.9956	0.9312	0.9941	0.9600	0.9962	0.9981
	PatternNet (IF=0.05)	0.9693	0.9968	0.9956	0.9719	0.9944	0.9689	0.9910	0.9909	0.9946	0.9992
	PatternNet (IF=0.1)	0.9500	0.9965	0.9962	0.9619	0.9984	0.9847	0.9951	0.9958	0.9964	0.9986

TABLE V
MAP RESULTS ON NWPURESISC45 LONG-TAILED RS DATASET WITH 16-, 32-, AND 64-BIT CODE LENGTH

	Methods	DHN	DSH	DTSH	DPSH	HashNet	FAH	DHCNN	AHCL	CSQ	CIAH (Ours)
16 bits	NWPURESISC45(IF=0.01)	0.7376	0.8851	0.8716	0.7759	0.8747	0.8297	0.6923	0.8014	0.8986	0.9485
	NWPURESISC45(IF=0.05)	0.6542	0.8824	0.8471	0.6522	0.8267	0.8511	0.7905	0.7135	0.9190	0.9429
	NWPURESISC45(IF=0.1)	0.6725	0.8860	0.8670	0.6010	0.7275	0.6852	0.7850	0.6609	0.8925	0.9488
32 bits	NWPURESISC45(IF=0.01)	0.8014	0.9053	0.8717	0.8215	0.9017	0.8403	0.7493	0.8925	0.9211	0.9560
	NWPURESISC45(IF=0.05)	0.7470	0.8983	0.8718	0.8249	0.9203	0.8636	0.8754	0.8693	0.9309	0.9508
	NWPURESISC45(IF=0.1)	0.7177	0.9164	0.8368	0.8176	0.8950	0.8690	0.8599	0.8699	0.9142	0.9526
64 bits	NWPURESISC45(IF=0.01)	0.8031	0.9028	0.8839	0.8740	0.9097	0.8495	0.7693	0.8643	0.9237	0.9610
	NWPURESISC45(IF=0.05)	0.7199	0.9218	0.8554	0.8507	0.9235	0.8707	0.8889	0.8413	0.9311	0.9591
	NWPURESISC45(IF=0.1)	0.6695	0.9216	0.8525	0.8491	0.9134	0.8676	0.8632	0.9003	0.9226	0.9587

(DSH) [50]; 2) deep hashing network (DHN) [51]; 3) HashNet [52]; 4) deep supervised hashing with triplet 930 labels (DTSH) [53]; 5) deep pairwise-supervised hashing (DPSH) [54]; 6) central similarity quantization (CSQ) [45]; 7) feature and hash (FAH) learning [35]; 8) deep hashing convolutional neural network (DHCNN) [36]; and 9) asymmetric hash code learning (AHCL) [37]. Among them, DSH, DHN, HashNet, DTSH, DPSH, and CSQ are representative hashing methods,

TABLE VI

ABLATION EXPERIMENTS OF THE PROPOSED CIAH WITH DIFFERENT PARAMETERS α . THE EXPERIMENTS ARE CONDUCTED ON CLRS (IF = 0.05) WITH 32-BIT CODE LENGTH

α	0.1	0.2	0.3	0.4	0.5
mAP	0.9261	0.9370	0.9331	0.9298	0.9275
α	0.6	0.7	0.8	0.9	1.0
mAP	0.9323	0.9287	0.9273	0.9313	0.9234

TABLE VII

ABLATION EXPERIMENTS OF THE PROPOSED CIAH. THE EXPERIMENTS ARE CONDUCTED ON CLRS (IF = 0.05) WITH 32-BIT CODE LENGTH

Component			Code length		
$\mathcal{L}_{C_1}$	$\mathcal{L}_{C_2}$	IAM	16 bits	32 bits	64 bits
✓			0.8720	0.8958	0.8444
✓	✓		0.9209	0.9217	0.9282
✓	✓	✓	0.9364	0.9370	0.9299

while FAH, DHCNN, and AHCL are competitive RS hashing methods developed in recent years. While only CSQ uses center loss, the remaining methods employ tuple similarity loss. Specifically, DTSH employs triplet loss, while others use pairwise loss. Although a variety of hash loss functions are available, pairwise loss is still mainstream. Extensive comparative experiments are conducted between the reference methods and the proposed CIAH approach on 12 long-tailed distribution RS datasets. The results are presented in the form of both numerical values and visualization figures.

1) *Ablation Study and Time Consuming*: In the global loss function, α has a significant impact. The investigation focuses on examining the effect on mAP when α is set to different values. Table VI shows the experimental results. According to it, the overall performance of the CIAH proposal is optimal when α is set to 0.2. As a result, α is set to 0.2 for all studies.

In addition, the effects of centripetal loss, classification loss, and the IAM are investigated by evaluating different combinations of centripetal loss $\mathcal{L}_{C_1}$, classification loss $\mathcal{L}_{C_2}$, and IAM. The results are summarized in Table VII. According to the results, $\mathcal{L}_{C_2}$ exerts a substantial effect on performance improvement. In addition, the inclusion of the IAM produces a slight increase in mAP. These experimental results support the favorable effect of each CIAH module on mAP enhancement.

Additionally, the complexity and time consumption of CIAH are examined. To train the model by minimizing or maximizing the distance between similar samples or dissimilar samples, traditional pairwise hash loss requires pairwise combinations of two out of N samples. This operation has a computational complexity of $O(N^2)$. In contrast, the centripetal loss involves only a one-by-one comparison of all samples with their respective centers, with a computational cost of $O(N)$. In terms of the introduced IAM, completely connected layers, activations, as well as multiplication and contact operations are employed. In conclusion, the evident outcome is the substantial boost in retrieval accuracy achieved by CIAH through the incorporation of subtle parameters.

2) *Overall Retrieval Performance*: Table II provides the mAP results on three shrinkage scale long-tailed datasets based on RSSCN7 with 16, 32, and 64 bits. Based on the obtained results, the proposed CIAH consistently achieves superior retrieval outcomes across most scenarios. Although CIAH is not the best at RSSCN7 (IF = 0.01) with 16 bits and RSSCN7 (IF = 0.05) with 32 bits, it achieves second-best performance in such cases. In a comprehensive assessment, summarizing the experimental results for the RSSCN7 dataset presents notable challenges. The dataset comprises a limited number of classes, and the samples within each class exhibit considerable dissimilarity. Consequently, these challenges are transferred to the model training, such as under-sampling of tail classes or over-fitting to head classes. Even within these challenging experimental conditions, CIAH demonstrates notable performance.

Table III provides the mAP results on three shrinkage scale long-tailed datasets based on CLRS with 16, 32, and 64 bits. We observe that our CIAH obtains the best mAP results. In contrast to the outcomes on the RSSCN7 dataset, the proposed CIAH exhibits a more pronounced enhancement in retrieval performance on the CLRS dataset. The average value of CIAH improved accuracy is 3.64% compared to the second-best method. This indicates a notable enhancement in terms of retrieval performance within the realm of RS datasets. In addition, the 32-bit-length hash code achieves the best retrieval performance. In the context of hash retrieval, no universally accepted criterion establishes a direct correlation between the length of the hash code and the subsequent improvement in retrieval performance. Therefore, in practical applications, it is critical to choose the proper length of the hash code.

Table IV provides the mAP results on three shrinkage scale long-tailed datasets based on PatternNet with 16, 32, and 64 bits. Similar to the previous dataset, it is observed that CIAH attains the highest retrieval performance although the enhancement is slight. The retrieval performance remains satisfactory across all the methods evaluated. This is due to the fact that the PatternNet dataset has large interclass differentiation. Although the PatternNet dataset contains 38 classes, the samples selected in each class have typical features of that class. In other words, images in the same class have a high similarity. Therefore, hashing can accurately distinguish the class to which the samples belong. This phenomenon results in the less pronounced performance enhancement of novel methods on this dataset.

Table V provides the mAP results on three shrinkage scale long-tailed dataset based on NWPURESISC45 with 16, 32, and 64 bits. The outcomes of the analysis indicate that the proposed CIAH not only attains the highest mAP results, but also manifests a substantial enhancement. Concretely, CIAH achieves an average accuracy improvement of 3.60%. Remarkably, within the complex context of the NWPURESISC45 dataset (encompassing 45 distinct land-use classes marked by pronounced imbalances in terms of sample quantities among classes), our CIAH model is still able to adapt to such a sophisticated experimental environment. This noteworthy performance unfolds despite the challenges posed by the demanding experimental conditions presented by this dataset.

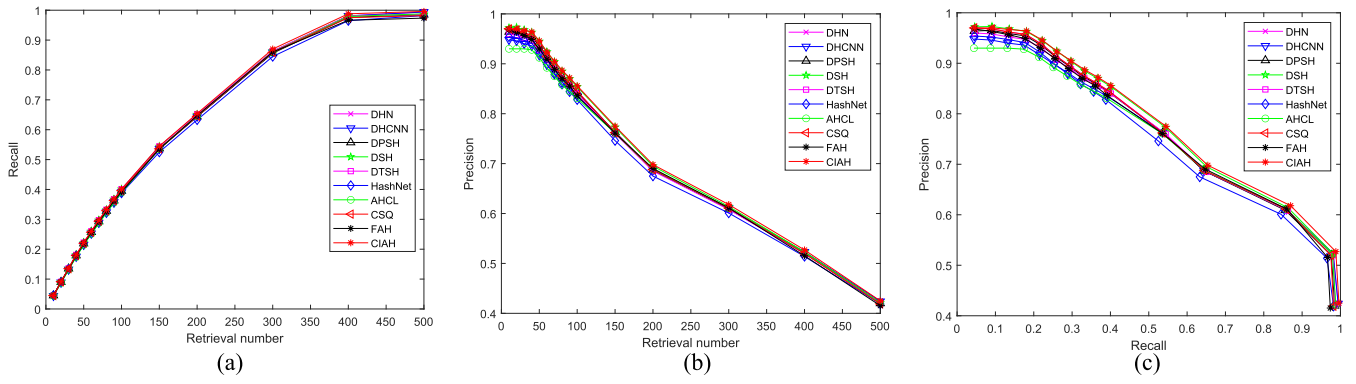


Fig. 7. (a) Recall-number, (b) precision-number, and (c) precision-recall curves on RSSCN7 (IF = 0.05) with 32-bit code length.

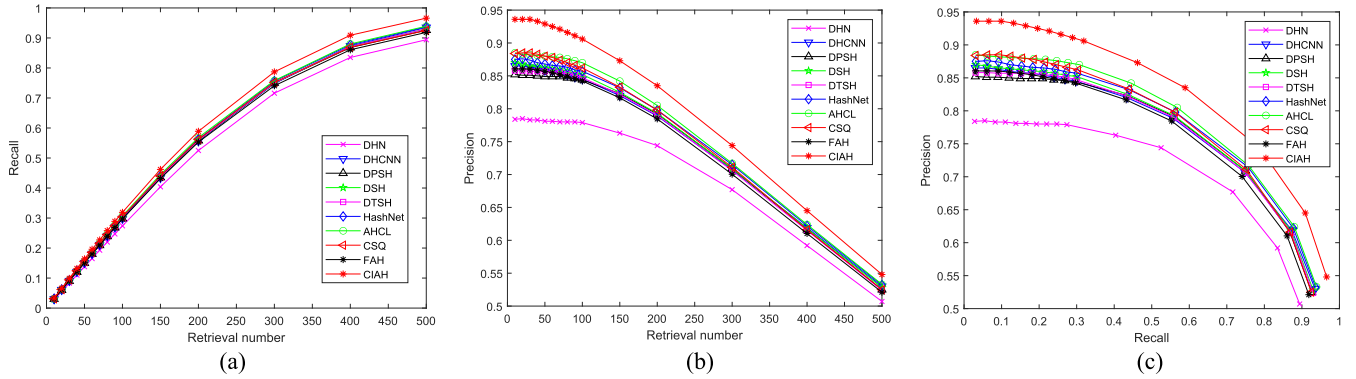


Fig. 8. (a) Recall-number, (b) precision-number, and (c) precision-recall curves on CLRS (IF = 0.05) with 32-bit code length.

Based on the ensemble of Tables II–V, a comprehensive analysis of the collective experimental outcomes is presented. Although there are three different IF values for the long-tailed distribution dataset, the retrieval performance is not sensitive to such IF values, except for the RSSCN7 dataset. This is because the RSSCN7 dataset has not enough samples, and different IF values have a strong impact on the volume of the dataset. As the IF value increases, the volume of the dataset also increases for deep hashing to have enough samples for hash code learning. In addition to these observations, it is notable that various loss functions exert an influence on the retrieval performance. Obviously, centripetal loss is more capable of generating discriminative hash codes on long-tailed distribution datasets compared to tuple similarity loss. This occurrence stems from the effectiveness of the centripetal loss in addressing the inherent imbalance within long-tailed distribution datasets. This, in turn, empowers the model to emphasize the tail classes to a significant degree. Also, CSQ employs centripetal loss, but the proposed CIAH still achieves better performance. The first reason is that CIAH uses an IAM to enhance the original features. This enhances features of the same symbol using a weight vector. Such operation facilitates the subsequent generation of discriminative hash codes. The second is that the hash code centers of CSQ are created in a hard-line manner, neglecting correlation between classes. In contrast, the generation of the proposed CIAH hash centers is data-driven. This results in a more reasonable hash code center. In summary, the two key strategies employed by

our CIAH, i.e., IAM and centripetal loss, jointly contribute to achieve very competitive retrieval performance.

3) *Assessment Through Precision-Recall Curves:* The precision-recall curves for the four RS datasets with IF = 0.05 and a hash code length of 32 bits are displayed in Figs. 7–10. Concretely, Figs. 7(a), 8(a), 9(a), and 10(a) provide the number of retrieval results returned (horizontal coordinate) and the recall (vertical coordinate). It reflects the percentage of correct samples recalled as the number of query results increases. The curve tends to be upward. Figs. 7(b), 8(b), 9(b), and 10(b) have the same horizontal coordinate of the number of retrieved results returned and the vertical coordinate of the precision. Moreover, it reflects the evolution of the precision as the number of query results increases. The curve usually trends downward. Lastly, Figs. 7(c), 8(c), 9(c), and 10(c) have the horizontal coordinate of recall and the vertical coordinate of precision. They reflect the relationship between recall and precision, which generally tends to be decreasing. The area enclosed by the precision recall curve and the axes are proportional to the retrieval performance. For a better visual presentation, some of the subfigures have been zoomed.

According to Figs. 7 and 9, the advantage of our CIAH is not evident in this context, which can be attributed to the scarcity of training samples for RSSCN7. This leads to an under-fitting profile during the training process. Furthermore, the PatternNet dataset consists of meticulously selected samples, each of which with the characteristic appearance of the class and not prone to hard identification. Fortunately,

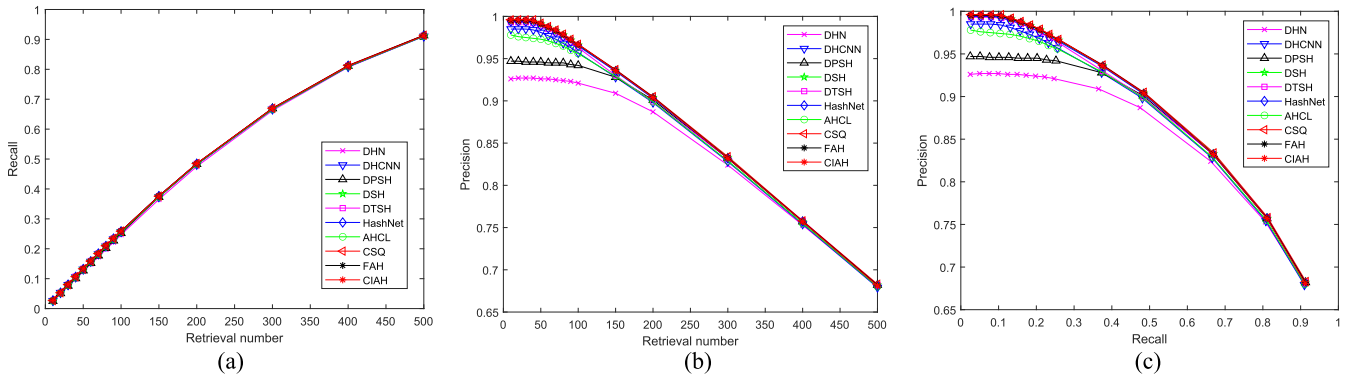


Fig. 9. (a) Recall-number, (b) precision-number, and (c) precision-recall curves on PatternNet (IF = 0.05) with 32-bit code length.

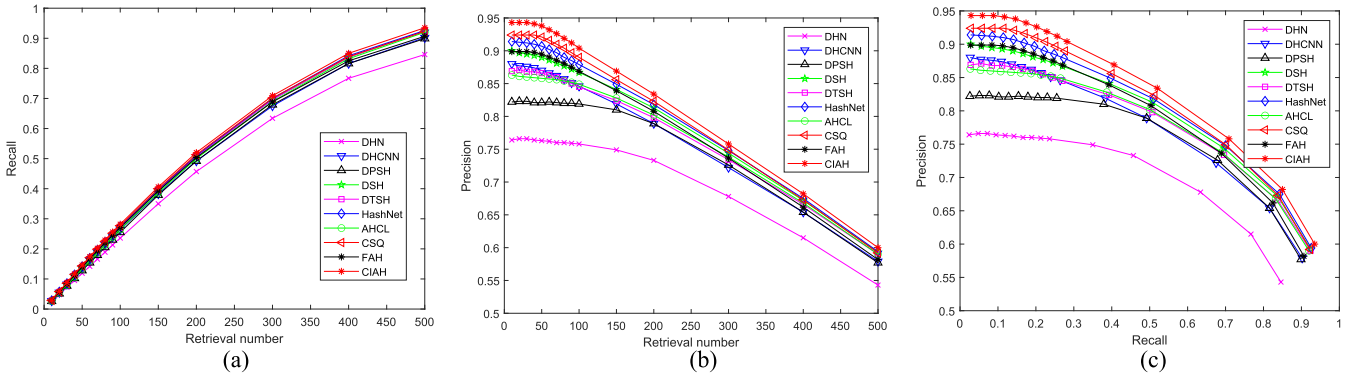


Fig. 10. (a) Recall-number, (b) precision-number, and (c) precision-recall curves on NWPURESISC45 (IF = 0.05) with 32-bit code length.

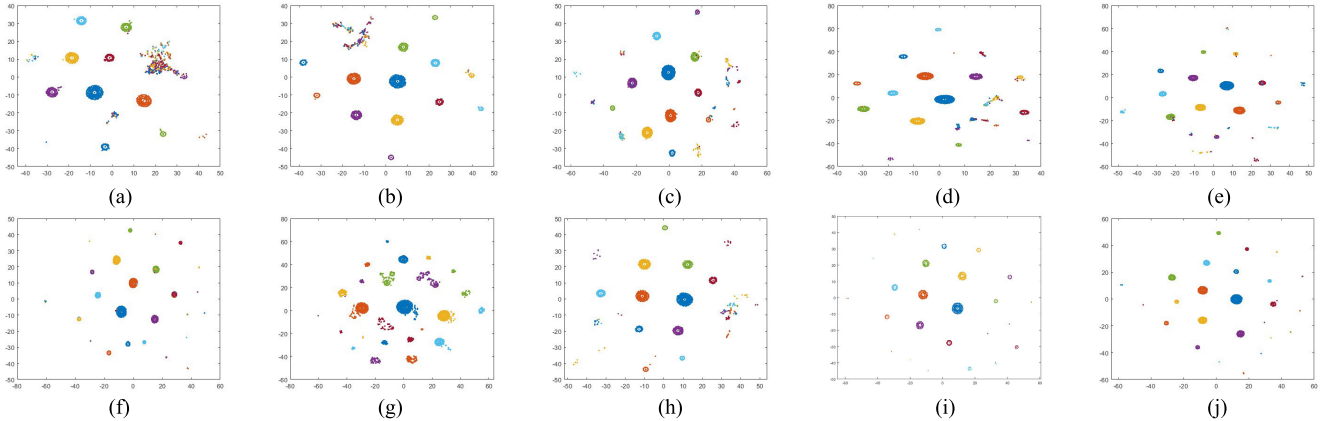


Fig. 11. t-SNE distribution of different hashing methods. The experiments are conducted on CLRS (IF = 0.05) with 32-bit code length. Different colored dots represent different classes of samples. (a) DHN. (b) DPSH. (c) DTSH. (d) DSH. (e) HashNet. (f) FAH. (g) DHCNN. (h) AHCL. (i) CSQ. (j) CIAH (Ours).

the results derived from the remaining two chosen datasets unequivocally exemplify the advantages of our CIAH model, as depicted in Figs. 8 and 10. Evidently, as the volume of retrieved results expands, the recall and precision curves of the CIAH approach consistently outperform those of all the compared methods. This indicates that CIAH retrieves more correct samples under the same conditions. This advantage is more evidently demonstrated in the recall-precision curve. With the increase of recall, the precision curve of CIAH decreases slowly although the trend is decreasing. Also, at all recall points, the precision of CIAH is significantly better than that of all comparison methods. All the visualization curve

results further demonstrate the superiority of the proposed CIAH and are consistent with the numerical results.

4) *t-SNE Visualization*: Illustrating an essential criterion for visual assessment in hash retrieval tasks, the outcomes of t-SNE analysis are presented in Fig. 11. Due to space limitation, the t-SNE results are provided for all methods at CLRS (IF = 0.05) with 32-bit hash code length. Evidently, the hash codes of all methods exhibit clustering. As previously mentioned, the objective of hashing is to bring hash codes of samples from the same class into proximity while ensuring that hash codes of samples from different classes remain distant. Thus, hash codes of the same class are clustered in

the Hamming space. Hash codes maintain the same clustering relationship in two dimensions by t-SNE dimensionality reduction. It can be discerned that hash codes generated through tuple similarity loss training tend to be clustered together. Nevertheless, a substantial proportion of sample points stray outside from the cluster centers. The rationale behind this observation lies in the fact that tuple similarity loss training exclusively accounts for the relationships among several samples, thereby neglecting the broader class relationships that encompass all the samples. Finally, the hash codes generated by CSQ and the proposed CIAH (both trained with centripetal loss) exhibit satisfactory clustering results. Obviously, the hash codes generated by CIAH are more cohesive. This phenomenon primarily stems from CIAH utilization of IAM to enhance features, resulting in features that possess heightened discriminatory qualities.

In summation, comprehensive experimentation substantiates that CIAH attains the forefront of performance in the domain of long-tailed distributed RS datasets. This affirmation underscores the model robustness and proficiency in effectively addressing the challenges posed by such datasets, affirming its efficacy in the context of retrieval tasks characterized by pronounced class imbalances.

IV. CONCLUSION

To complement the gap in fast retrieval of long-tailed distributed RS datasets, this article proposes a novel centripetal hashing model for effective long-tailed RS image retrieval (named CIAH), which is based on intensive attention features. Such long-tailed distributed datasets are imbalanced compared to regular datasets. To address this drawback, an end-to-end deep learning model has been developed to generate discriminative hash codes. Specifically, an IAM is employed to generate attention vectors, thereby enhancing the representative features. Such features notably contribute to the differentiation of the subsequently generated hash codes. Furthermore, a centripetal loss is adopted to optimize the parameters of the model. First, the hash code class centers are predefined and the centripetal loss is leveraged to make all hash codes converge toward their corresponding hash class centers. This alleviates the attention bias problem, which occurs when the popular tuple similarity loss is applied to long-tailed distribution datasets. The above two strategies collectively contribute to ensuring that the proposed CIAH is ideally adaptable to such scenarios. To finalize, a comprehensive set of long-tailed distributed RS datasets has been meticulously constructed, subsequently releasing them as a publicly accessible benchmark to serve as a standardized reference for this task. The extensive and all-encompassing experimental analyses carried out corroborate the compelling efficacy of the newly proposed CIAH model, thereby reaffirming its prowess in the domain of image retrieval characterized by pronounced class imbalances.

REFERENCES

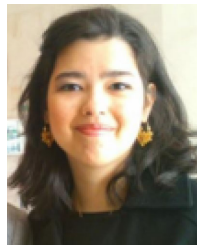
- [1] USGS and NASA. (2021). *Earth Observing 1, Hyperion Website*. [Online]. Available: <https://eo1.usgs.gov/sensors/hyperion>
- [2] B. Zhang et al., "Progress and challenges in intelligent remote sensing satellite systems," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1814–1822, 2022.
- [3] S. Pignatti et al., "The PRISMA hyperspectral mission: Science activities and opportunities for agriculture and land monitoring," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2013, pp. 4558–4561.
- [4] K. E. Joyce, S. E. Belliss, S. V. Samsonov, S. J. McNeill, and P. J. Glassey, "A review of the status of satellite remote sensing and image processing techniques for mapping natural hazards and disasters," *Prog. Phys. Geography, Earth Environ.*, vol. 33, no. 2, pp. 183–207, Apr. 2009.
- [5] H. S. Munawar, A. W. A. Hammad, and S. T. Waller, "Remote sensing methods for flood prediction: A review," *Sensors*, vol. 22, no. 3, p. 960, Jan. 2022.
- [6] R. T. Eguchi, C. K. Huyck, S. Ghosh, and B. J. Adams, "The application of remote sensing technologies for disaster management," in *Proc. 14th World Conf. Earthq. Eng.*, vol. 17, Oct. 2008.
- [7] Y. Xiao and Q. Zhan, "A review of remote sensing applications in urban planning and management in China," in *Proc. Joint Urban Remote Sens. Event*, May 2009, pp. 1–5.
- [8] T. Wellmann et al., "Remote sensing in urban planning: Contributions towards ecologically sound policies?" *Landscape Urban Planning*, vol. 204, Dec. 2020, Art. no. 103921.
- [9] N. Hoalst-Pullen and M. W. Patterson, "Applications and trends of remote sensing in professional urban planning," *Geography Compass*, vol. 5, no. 5, pp. 249–261, May 2011.
- [10] P. K. Mallupattu et al., "Analysis of land use/land cover changes using remote sensing data and GIS at an urban area, Tirupati, India," *Sci. World J.*, vol. 2013, May 2013, Art. no. 268623.
- [11] X.-L. Chen, H.-M. Zhao, P.-X. Li, and Z.-Y. Yin, "Remote sensing image-based analysis of the relationship between urban heat island and land use/cover changes," *Remote Sens. Environ.*, vol. 104, no. 2, pp. 133–146, Sep. 2006.
- [12] J. Jiang and G. Tian, "Analysis of the impact of land use/land cover change on land surface temperature with remote sensing," *Proc. Environ. Sci.*, vol. 2, pp. 571–575, Jan. 2010.
- [13] A. Swaminathan, Y. Mao, and W. Min, "Robust and secure image hashing," *IEEE Trans. Inf. Forensics Security*, vol. 1, no. 2, pp. 215–230, Jun. 2006.
- [14] J. Shin and C. Ruland, "A survey of image hashing technique for data authentication in WMSNs," in *Proc. IEEE 9th Int. Conf. Wireless Mobile Comput., Netw. Commun. (WiMob)*, Oct. 2013, pp. 253–258.
- [15] B. Kulis and T. Darrell, "Learning to hash with binary reconstructive embeddings," in *Proc. NIPS*, vol. 22, 2009, pp. 1042–1050.
- [16] J. Wang, T. Zhang, J. Song, N. Sebe, and H. T. Shen, "A survey on learning to hash," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 769–790, Apr. 2018.
- [17] X. Shen, Y. Tang, Y. Zheng, Y.-H. Yuan, and Q.-S. Sun, "Unsupervised multiview distributed hashing for large-scale retrieval," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 12, pp. 8837–8848, Dec. 2022.
- [18] L. Han, P. Li, X. Bai, C. Grecos, X. Zhang, and P. Ren, "Cohesion intensive deep hashing for remote sensing image retrieval," *Remote Sens.*, vol. 12, no. 1, p. 101, Dec. 2019.
- [19] P. Li et al., "Hashing nets for hashing: A quantized deep learning to hash framework for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7331–7345, Oct. 2020.
- [20] Y. Fang, P. Li, J. Zhang, and P. Ren, "Cohesion intensive hash code book coconstruction for efficiently localizing sketch depicted scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5629016.
- [21] J. Gao, X. Shen, P. Fu, Z. Ji, and T. Wang, "Multiview graph convolutional hashing for multisource remote sensing image retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 8015305.
- [22] Y. Sun, W. Wu, X. Shen, and Z. Cui, "Multiview inherent graph hashing for large-scale remote sensing image retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10705–10715, 2021.
- [23] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.
- [24] A. Oliva and A. Torralba, "Modeling the shape of the scene: A holistic representation of the spatial envelope," *Int. J. Comput. Vis.*, vol. 42, no. 3, pp. 145–175, 2001.
- [25] B. Demir and L. Bruzzone, "Hashing-based scalable remote sensing image search and retrieval in large archives," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 2, pp. 892–904, Feb. 2016.

- [26] P. Li and P. Ren, "Partial randomness hashing for large-scale remote sensing image retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 3, pp. 464–468, Mar. 2017.
- [27] T. Reato, B. Demir, and L. Bruzzone, "An unsupervised multicode hashing method for accurate and scalable remote sensing image retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 276–280, Feb. 2019.
- [28] R. Fernandez-Beltran, B. Demir, F. Pla, and A. Plaza, "Unsupervised remote sensing image retrieval using probabilistic latent semantic hashing," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 2, pp. 256–260, Feb. 2021.
- [29] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
- [30] Y. Kim, "Convolutional neural networks for sentence classification," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*. Association for Computational Linguistics, 2014.
- [31] A. Karpathy, G. Toderici, S. Shetty, T. Leung, R. Sukthankar, and L. Fei-Fei, "Large-scale video classification with convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1725–1732.
- [32] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How transferable are features in deep neural networks?" in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 27, 2014.
- [33] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3D convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2015, pp. 4489–4497.
- [34] Y. Li, Y. Zhang, X. Huang, H. Zhu, and J. Ma, "Large-scale remote sensing image retrieval by deep hashing neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 950–965, Feb. 2018.
- [35] C. Liu, J. Ma, X. Tang, F. Liu, X. Zhang, and L. Jiao, "Deep hash learning for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3420–3443, Apr. 2021.
- [36] W. Song, S. Li, and J. A. Benediktsson, "Deep hashing learning for visual and semantic retrieval of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9661–9672, Nov. 2021.
- [37] W. Song, Z. Gao, R. Dian, P. Ghamisi, Y. Zhang, and J. A. Benediktsson, "Asymmetric hash code learning for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5617514.
- [38] L. Han, P. Li, A. Plaza, and P. Ren, "Hashing for localization (HfL): A baseline for fast localizing objects in a large-scale scene," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609916.
- [39] J. Han, P. Li, Y. Tao, and P. Ren, "Encrypting hashing against localization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5607414.
- [40] T. Wu, Q. Huang, Z. Liu, Y. Wang, and D. Lin, "Distribution-balanced loss for multi-label classification in long-tailed datasets," in *Proc. 16th Eur. Conf. Comput. Vis. Glasgow, U.K.: Springer*, 2020, pp. 162–178.
- [41] P. Chu, X. Bian, S. Liu, and H. Ling, "Feature space augmentation for long-tailed data," in *Proc. 16th Eur. Conf. Comput. Vis. Glasgow, U.K.: Springer*, 2020, pp. 694–710.
- [42] T. Wang, Y. Zhu, C. Zhao, W. Zeng, J. Wang, and M. Tang, "Adaptive class suppression loss for long-tail object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 3103–3112.
- [43] J. Tan et al., "Equalization loss for long-tailed object recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11662–11671.
- [44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [45] L. Yuan et al., "Central similarity quantization for efficient image and video retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3083–3092.
- [46] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.
- [47] P. Veličković et al., "The CLRS algorithmic reasoning benchmark," 2022, *arXiv:2205.15659*.
- [48] W. Zhou, S. Newsam, C. Li, and Z. Shao, "PatternNet: A benchmark dataset for performance evaluation of remote sensing image retrieval," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 197–209, Nov. 2018.
- [49] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.
- [50] H. Liu, R. Wang, S. Shan, and X. Chen, "Deep supervised hashing for fast image retrieval," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2064–2072.
- [51] H. Zhu, M. Long, J. Wang, and Y. Cao, "Deep hashing network for efficient similarity retrieval," in *Proc. 13th AAAI Conf. Artif. Intell.*, 2016, pp. 2415–2421.
- [52] Z. Cao, M. Long, J. Wang, and P. S. Yu, "HashNet: Deep learning to hash by continuation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 5608–5617.
- [53] X. Wang, S. Yi, and K. M. Kitani, *Deep Supervised Hashing With Triplet Labels*. Cham, Switzerland: Springer, 2016.
- [54] W. Li, S. Wang, and W. Kang, "Feature learning based deep supervised hashing with pairwise labels," in *Proc. Int. Joint Conf. Artif. Intell.*, 2016, pp. 1711–1717.



Lirong Han (Student Member, IEEE) received the B.Eng. degree from the Qingdao University of Science and Technology, Qingdao, China, in 2017, and the M.Eng. degree in information and communication engineering from the China University of Petroleum (East China), Qingdao, in 2020. She is currently pursuing the Ph.D. degree with the Department of Technology of Computers and Communications, University of Extremadura, Cáceres, Spain.

Her research interests include deep learning and hash-based retrieval method, with applications to remote sensing.



Mercedes E. Paoletti (Senior Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from the University of Extremadura, Cáceres, Spain, in 2014, 2016, and 2020, respectively.

She is currently an Associate Professor with the Mérida University Centre, Department of Technology of Computers and Communications, University of Extremadura, and also a member of the Hyper-spectral Computing Laboratory (HyperComp). Her research interests include remote sensing analysis through deep learning models and high-performance

computing.

Dr. Paoletti was a recipient of the 2019 Outstanding Paper Award Recognition in the IEEE WHISPERS 2019 Conference, the Outstanding Ph.D. Award at the University of Extremadura in 2020, and the Research Award from the Spanish Scientific Society of Computer Science and the BBVA Foundation in 2022. She has been a reviewer for different journals and congresses, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL), an activity for which she has been recognized as best reviewer in 2019 and 2020. She has also guest edited four special issues on hyperspectral data and deep learning analysis for *Remote Sensing*, and served as a Topical Associate Editor for IEEE TGRS from 2021 to 2023.

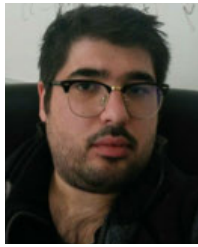


Sergio Moreno-Álvarez (Graduate Student Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in computer engineering from the University of Extremadura, Cáceres, Spain, in 2017, 2019, and 2022, respectively.

He is currently a member and a Professor with the Department of Computer Systems and Languages, National University of Distance Education (UNED), Madrid, Spain. He has authored or coauthored more than 15 JCR journal articles and multiple presentations at international and national conferences. His

research interests cover high-performance computing (HPC), deep learning, and its optimization through parallelism approaches and remote sensing (RS).

Dr. Moreno-Álvarez has served as a reviewer for multiple IEEE journals, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTATIONAL INTELLIGENCE, and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Juan M. Haut (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in computer engineering and the Ph.D. degree in information technology from the University of Extremadura, Cáceres, Spain, in 2011, 2014, and 2019, respectively, through the support by the University Teacher Training Programme from the Spanish Ministry of Education.

He is currently a Professor with the Department of Computers and Communications, University of Extremadura, where he is also a member of the Hyperspectral Computing Laboratory (HyperComp), Department of Technology of Computers and Communications. His research interests include remote sensing data processing and high-dimensional data analysis, applying machine (deep) learning, and cloud computing approaches. In this sense, he has authored or coauthored more than 65 JCR journal articles (more than 30 in IEEE journals) and more than 30 peer-reviewed conference proceeding papers. Some of his contributions have been recognized as hot-topic publications for their impact on the scientific community.

Dr. Haut was a recipient of the Outstanding Ph.D. Award at the University of Extremadura in 2019, and the Outstanding Paper Award in the 2019 and 2021 IEEE WHISPERS Conferences. From his experience as a reviewer, it is worth mentioning his active collaboration in more than ten scientific journals, such as IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL). He has been awarded with the Best Reviewer Recognition of IEEE GRSL and IEEE TGRS, in 2018 and 2020, respectively. Furthermore, he has guest-edited three special issues on hyperspectral remote sensing for different journals. He is also an Associate Editor of IEEE GRSL and IEEE JOURNAL ON MINIATURIZATION FOR AIR AND SPACE SYSTEMS.



Rafael Pastor-Vargas (Senior Member, IEEE) received the M.Sc. degree in physics from Complutense University, Madrid, Spain, in 1994, and the Ph.D. degree in computer science from the Universidad Nacional de Educación a Distancia (UNED), Madrid, in 2006.

He was an Innovation Manager with the Innovation and Development Centre, UNED, from 2004 to 2009, and the General Manager for incorporating innovative services to UNED's learning model from 2009 to 2011. He is currently an Associate Professor with the Control and Communication Systems Department, UNED, where he is also the Dean of the Computer Science Engineering Faculty. He has participated in a big number of projects financed in public calls, some of them with special relevance for companies and administrations with an international scope, international journals, and

conferences. He teaches graduate and postgraduate courses related to network interconnection, security domains, and cloud computing, among others. His research interests include the quality of service support in distributed systems, applied machine learning algorithms, and the development of infrastructure and algorithms for e-learning, remote sensing, and cybersecurity, particularly on the Cloud.

Dr. Pastor-Vargas belongs to the Blockchain Local Group of the IEEE Spain Section and is also vocal in the IEEE Education Society (Spanish Chapter).



Antonio Plaza (Fellow, IEEE) received the M.Sc. and Ph.D. degrees in computer engineering from the University of Extremadura, Badajoz, Spain, in 1999 and 2002, respectively.

He is currently a Full Professor and the Head of the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura, Cáceres, Spain. His research interests include hyperspectral data processing and parallel computing of remote sensing data.

Dr. Plaza was a recipient of the Recognition of Best Reviewers of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2009 and the Recognition of Best Reviewers of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) in 2010, for which he served as an Associate Editor, from 2007 to 2012. He was a recipient of the Best Column Award of *IEEE Signal Processing Magazine* in 2015, the 2013 Best Paper Award of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS), and the most highly cited article from 2005 to 2010 in the *Journal of Parallel and Distributed Computing*. He received best paper awards at the IEEE Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing and the IEEE Symposium on Signal Processing and Information Technology. He is a member of Academia Europaea, The Academy of Europe. He was also a member of the Steering Committee of IEEE JSTARS. He served as the Director of the Education Activities for the IEEE Geoscience and Remote Sensing Society (GRSS) from 2011 to 2012 and the President of the Spanish Chapter for IEEE GRSS from 2012 to 2016. He is currently serving as the Chair for the Publications Awards Committee of the IEEE GRSS and the Vice-Chair for the Fellow Evaluations Committee of the IEEE GRSS. He served as the Editor-in-Chief for IEEE TGRS from 2013 to 2017. He is currently serving as the Editor-in-Chief for IEEE JOURNAL OF MINIATURIZED AIR AND SPACE SYSTEMS. He has been included in the 2018, 2019, 2020, and 2021 Highly Cited Researchers List (Clarivate Analytics).